KUYFI

BLACK-BOX SECURITY REPORT

Before the Breach.

Contract ID	CCOIK76OU7VHRK5PCE3RKQUDO6C6I7YMBHM3G24FVMZJ2I2ZESBGPHYZ
Network	TESTNET
Report ID	kyf-20261025T120251Z-9974ec83
Generated At	2026-10-25T12:02:51.753Z
Tool Version	Kuyfi 0.1.0
Schema Version	1.0.0

This report is generated by automated black-box fuzzing against a deployed Soroban smart contract. It is not a manual audit and does not constitute a formal security certification.

2. Executive Summary

Functions detected	5
UDTs detected	3
Vectors executed	48
Broadcast transactions	0
Transactions with hash	0
Total findings	0

Findings by Severity

Critical	0
High	0
Medium	0
Low	0
Info	0

3. Attack Surface

Target

Contract ID	CCOIK76OU7VHRK5PCE3RKQUDO6C6I7YMBHM3G24FVMZJ2I2ZESBGPHYZ
Network	TESTNET
Bytecode size	7905 bytes

Functions Detected (5)

- `init(admin: Address, verifier: Address) -> returns value`
- `claim(chest_id: U64, recipient: Address, nullifier: BytesN<32>, proof: Proof) -> returns value`
- `get_chest(chest_id: U64) -> returns value`
- `close_chest(chest_id: U64) -> returns value`
- `create_chest(creator: Address, fingerprint: BytesN<32>, token: Address, prize: I128, max_claims: U32, deadline: U64) -> returns value`

UDTs Detected (3)

STRUCT Chest

- `claims: U32`
- `closed: Bool`
- `creator: Address`
- `deadline: U64`
- `fingerprint: BytesN<32>`
- `max_claims: U32`
- `prize: I128`
- `token: Address`

STRUCT Proof

- `a: BytesN<64>`
- `b: BytesN<128>`
- `c: BytesN<64>`

UNION DataKey

- `Verifier`
- `NextId`
- `Chest(U64)`
- `Nullifier(BytesN<32>)`

4. Chaos Execution

Kuyfi performs black-box fuzzing against the deployed Soroban contract using its on-chain contract specification.

Vectors executed	48
Broadcast transactions	0
Transactions with hash	0
Verification transactions	0

5. Findings

NO REPORTABLE FINDINGS

No reportable security findings were produced by this audit run. This does not constitute a formal proof that the contract is vulnerability-free.

6. Verification

Real Testnet transactions broadcast during this audit run, provided so every claim above can be independently verified on-chain.

No transaction ever reached broadcast during this run.